Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Subhajeet Rath	**Email**	subhajeet.rath@tno.nl
Institute Name	Nederlandse Organisatie voor Toegepast Natuurwetenschappelijk Onderzoek (TNO), Powertrains Department			
Address	Automotive Campus 30, 5708 JZ Helmond, The Netherlands			

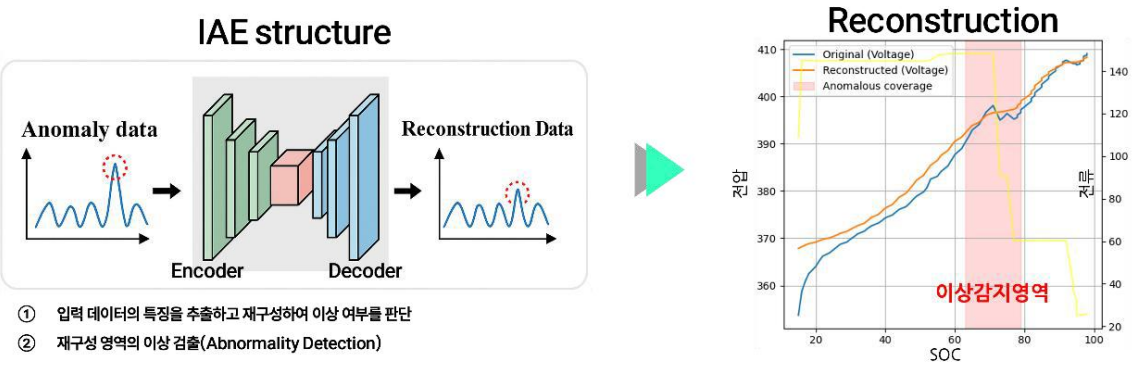
2. Technology Overview

Title of Proposed Technology	AI-based early detection and warning system for preventing battery fire (Interpretable AutoEncoder-based anomaly diagnostics on BMS and charging data)			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-generation Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-generation Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification 1)	Technology Field 1 (Large Category)		Technology Field 2 (Large Category)	
	Electrical & Electronics		Information & Communications	
	Medium Category	Small Category	Medium Category	Small Category
	Instrumentation	Safety monitoring/diagnosis measurement and control instruments	Software	System Integration

Concept of Proposed Technology

The proposed technology focuses on the development and validation of an AI-based early detection and warning system to prevent battery fire and thermal runaway in electric vehicle (EV) and industrial battery systems. The system targets early-stage abnormal behaviors that precede safety-critical events, using operational data already available from Battery Management Systems (BMS) and charging infrastructure.

The core technical approach is based on an Interpretable AutoEncoder (IAE) for unsupervised anomaly detection. The IAE is trained exclusively on normal-operation multivariate time-series data, including voltage, current, state of charge (SOC), and temperature signals where available. By learning compact latent representations of healthy battery behavior, the model establishes a data-driven baseline that captures normal electro-thermal and charging dynamics. [1]



During real-time operation, incoming data streams are reconstructed by the IAE, and reconstruction error metrics (e.g., RMSE across signals and time windows) are continuously evaluated. Deviations from learned normal patterns are translated into anomaly scores, which indicate potential precursors to abnormal battery states such as internal resistance increase, abnormal charging behavior, accelerated aging, or early thermal instability.

A key differentiating feature of the proposed technology is its emphasis on interpretability and explainability, which is critical for safety-related applications. The IAE framework is designed to provide signal-level contribution indicators, allowing identification of which variables most strongly drive detected anomalies. Temporal attribution further distinguishes between transient disturbances and persistent degradation patterns, enabling engineers and operators to understand why an alert is generated and to take informed preventive actions.

The anomaly scores are mapped to a multi-level early-warning logic, typically comprising normal, warning, and alarm states. Adaptive thresholds are applied to account for operating context (e.g., charging conditions, battery usage patterns, and aging effects), with the goal of achieving early warning hours to days ahead of potential safety incidents while maintaining a low false-alarm rate.

The system is designed for practical deployment in fleet and charging environments, supporting both cloud-based and edge-based architectures. Key components include secure data ingestion,

	real-time preprocessing and feature engineering, continuous anomaly scoring, and a monitoring dashboard that visualizes risk levels, anomaly trends, and interpretable indicators at battery, vehicle, and fleet scales. This architecture supports scalability and aligns with a battery safety management SaaS commercialization pathway. Within the joint R&D framework, the technology will be validated using a combination of field operational data and controlled experimental datasets, including abnormal scenarios such as accelerated aging, resistance growth, and abnormal charging. This ensures that the AI models are not only statistically effective but also physically meaningful and robust across operating conditions. Overall, the proposed technology integrates unsupervised AI-based anomaly detection, engineering-oriented interpretability, and deployment-ready system design to enable proactive battery fire risk prevention in next-generation mobility and energy systems. [2-3]
Rationale for Joint R&D Project	Battery fires are rare but high-impact events, and early detection from operational data remains challenging due to complex failure mechanisms and limited fault labels. [3] Recent peer-reviewed work demonstrates that unsupervised autoencoder approaches can detect battery abnormalities from real EV charging/BMS data using reconstruction error (RMSE) while enabling extensibility across vehicles. [2] This joint R&D combines Korean SME's interpretable autoencoder anomaly-diagnosis approach and field deployment know-how [1] with TNO's battery test and validation infrastructure and advanced BMS and diagnostics expertise on BMS technologies, safety, and fault diagnostics. [4–8]
Key Words	1. Battery safety / thermal runaway early warning 2. Interpretable AutoEncoder (IAE) 3. Unsupervised time-series anomaly detection 4. BMS and charging infrastructure data 5. Real-time monitoring dashboard / SaaS

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) 2)	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	1) Build representative normal-operation and abnormal-scenario datasets (including thermal runaway precursors, accelerated aging, and resistance increase cases) using a combination of field data and controlled experiments. [1,6,7] 2) Develop an Interpretable AutoEncoder-based anomaly detection model with reconstruction-error scoring and explainability outputs (variable contribution/pattern attribution). [1,2] 3) Implement a real-time early warning pipeline (edge or cloud) and visualization dashboard for operators and fleet managers. [1] 4) Validate performance and robustness across operating conditions and propose a commercialization pathway (battery safety management SaaS). [1,4,6]			
R&D Target Specifications	Target specifications will be finalized in Phase 1 based on accessible datasets and stakeholder requirements. Initial KPI targets (illustrative): • Detection lead time: early warning hours-to-days ahead of safety-critical events (where ground truth exists) [3] • False alarm rate: ≤ 1 per 1,000 charging sessions (to be validated per fleet) • Real-time latency: ≤ 5 minutes from data arrival to risk-score update • Explainability: ranked contributing signals/features for each alert (to support decision making) [2] • Deployment: scalable pipeline supporting fleet-scale data ingestion and monitoring dashboard. [1]			
R&D Plan	Total project duration: up to 36 months (aligned with TIPA Global Cooperation R&D Program). **WP1 – Requirements, Data, and Baseline Setup** Duration: Month 0–6 Main Activities: • Define system requirements and KPI targets for early battery fire detection • Establish data access, governance, and security framework • Preprocess and synchronize BMS and charging time-series data • Perform baseline feature engineering and data quality assessment • Construct representative normal-operation datasets			

WP2 – Interpretable AutoEncoder Model Development

Duration: Month 6–15

Main Activities:

- Design Interpretable AutoEncoder architecture for multivariate battery data
- Train models using normal-operation datasets only
- Implement reconstruction-error–based anomaly scoring
- Develop signal-level contribution and temporal attribution method
- Conduct offline validation and initial threshold definition

WP3 – Abnormal Scenario Generation and Model Refinement

Duration: Month 12–24

Main Activities:

- Design and execute controlled battery tests (e.g., accelerated aging, resistance increase, abnormal charging)
- Generate and curate abnormal-scenario datasets with metadata
- Integrate experimental and field data for model enhancement
- Refine adaptive thresholds and context-aware risk scoring
- Evaluate robustness across operating and aging conditions

WP4 – Real-Time Pipeline and Monitoring Dashboard

Duration: Month 18–30

Main Activities:

- Develop real-time data ingestion and processing pipeline (edge/cloud)
- Integrate anomaly detection and risk scoring modules
- Implement monitoring dashboard with interpretable indicators and alerts
- Perform system integration, scalability, and latency testing

WP5 – Pilot Demonstration and Commercialization Assessment

Duration: Month 24–36

Main Activities:

- Deploy system in pilot operational environments
- Validate KPIs (detection lead time, false-alarm rate, latency, explainability)
- Conduct reliability testing aligned with TRL 7
- Assess commercialization potential and SaaS deployment strategy

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO responsibilities: • Design and execute controlled battery experiments to generate labeled abnormal scenarios (e.g., accelerated aging, resistance increase, abnormal charging) and provide high-quality measurement protocols and metadata. [6,7] • Provide battery test infrastructure and know-how (cell testing, EIS where applicable, module/pack environmental testing) to validate model robustness under varying conditions. [6,7] • Contribute BMS and diagnostics expertise (fault diagnostics and safety considerations) and support performance evaluation and validation methodology (**Error! Reference source not found.**). [4] • Support EU-facing dissemination, potential standardization inputs, and business exploration for safety monitoring services. [6]
Expected R&R of Korean SME Partner	Korean SME's responsibilities: • Provide existing IAE anomaly diagnosis technology and implement the data processing pipeline (ingest, preprocessing, feature engineering). [1] • Develop, train, and tune the IAE-based anomaly detection and risk scoring algorithms based on RMSE-based anomaly score, adaptive thresholds, and interpretable outputs (**Error! Reference source not found.**). [1,2] • Build the real-time monitoring dashboard and integrate the model with charging/fleet data sources; conduct pilot operation and iterate based on feedback. [1] • Lead commercialization feasibility assessment for a battery safety management SaaS offering and support technology transfer activities.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes and impact: • Validated an interpretable AutoEncoder-based anomaly detection model and early-warning logic for battery fire risk. • Curated datasets and evaluation protocol for battery anomaly detection across operating conditions. • Prototype real-time monitoring dashboard suitable for fleet and charging-operator use (towards SaaS). • Improved battery safety management capability, with potential IP, publications, and international standardization contributions.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization